

Problem

Design an analog computer circuit to solve for $v(t)$, given the following equation and a value for $f(t)$ and that $v(0) = 0$ V.

$$(dv(t)/dt) + 3vt = ftdt$$

Step-by-step solution

Step 1 of 5

The differential equation for designing the analog computer is,

$$\frac{dv(t)}{dt} + 3v(t) = f(t)$$

Rewrite the above equation.

$$\frac{dv(t)}{dt} = f(t) - 3v(t) \dots\dots (1)$$

Step 2 of 5

If the input to the analog computer is $\frac{dv(t)}{dt}$, then the output of the integrator is $-v(t)$ as shown in Figure 1.

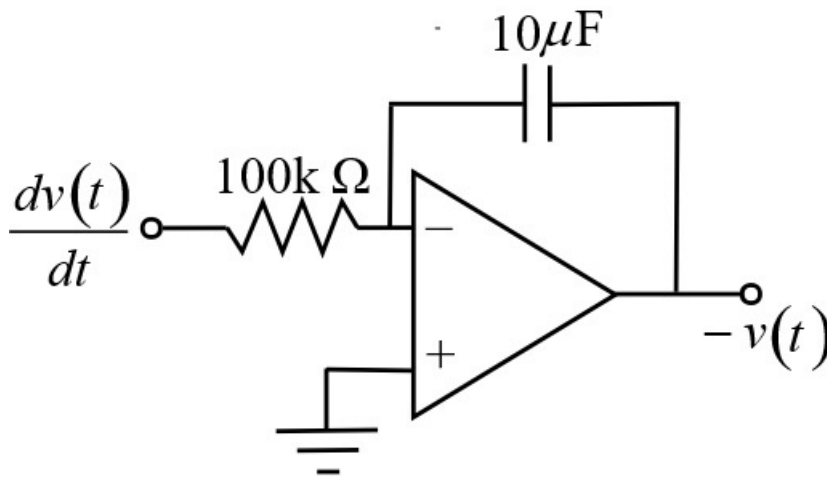


Figure 1

Step 3 of 5

To obtain the equation $f(t) - 3v(t)$, draw a summing amplifier.

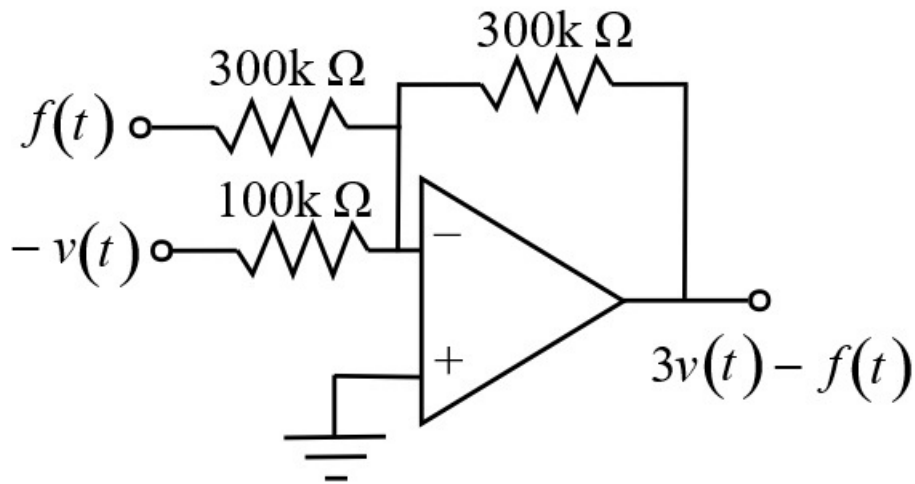


Figure 2

Step 4 of 5

Draw the inverting amplifier to obtain equation $f(t) - 3v(t)$.

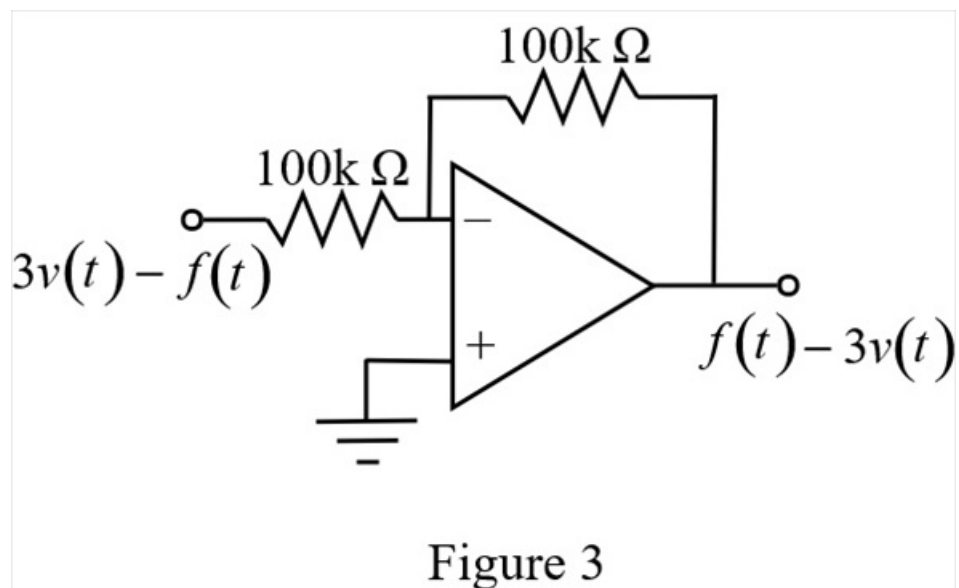


Figure 3

Step 5 of 5

Combine the Figure 1, 2 and 3 to obtain the equation (1) as shown in Figure 4.

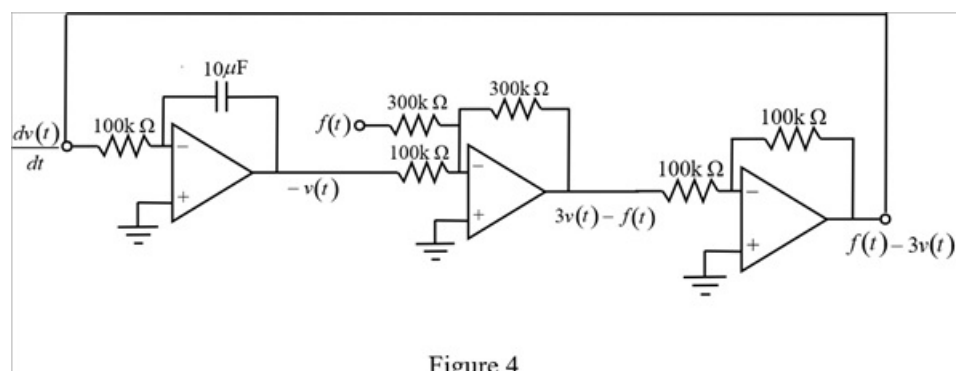


Figure 4

Thus, an analog computer circuit is designed.

